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(54) **PROCESS FOR THE PREPARATION OF SUGAMMADEX**

(71) Applicant: **FRESENIUS KABI IPSUM S.R.L.**, Cassina de' Pecchi (Milan) (IT)

(72) Inventors: **Walter Cabri**, Milan (IT); **Antonio Ricci**, Pescara (IT); **Jacopo Zanon**, Venice (IT); **Saswata Lahiri**, Ghaziabad (IN); **Govind Singh**, Ghaziabad (IN); **Shivaji Haribhau Shelke**, Maharashtra (IN); **Tapanjyoti Biswal**, Haryana (IN); **Nitin Kumar**, Delhi (IN); **Madan Singh**, Delhi (IN)

(73) Assignee: **Fresenius Kabi iPSUM S.r.l.**, Cassina de' Pecchi (Milan) (IT)

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C08B 37/00	(2006.01)

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CPC ***C08B 37/0012*** (2013.01); ***C07C 319/12*** (2013.01); ***C07C 323/52*** (2013.01); ***C08B 37/0003*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Layla D Berry
(74) *Attorney, Agent, or Firm* — Leydig, Voit & Mayer, Ltd.

(57) **ABSTRACT**

The present application provides an improved process for the preparation of sugammadex by reacting a salt of 3-mercapto propionic acid with 6-per-deoxy-6-per-halo-γ-cyclodextrin in a suitable organic solvent. This application also provides crystalline form of a salt of 3-mercapto propionic acid, preferably di sodium salt of 3-mercapto propionic acid and its use for the preparation of sugammadex, whereas formula (I) shows the sodium salt of sugammadex.

17 Claims, 1 Drawing Sheet

PROCESS FOR THE PREPARATION OF SUGAMMADEX

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application is the U.S. national stage of International Patent Application No. PCT/IB2017/051594, filed Mar. 20, 2017, which claims the benefit of Indian Patent Application No. IN201611009993, filed Mar. 22, 2016, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

FIELD OF THE INVENTION

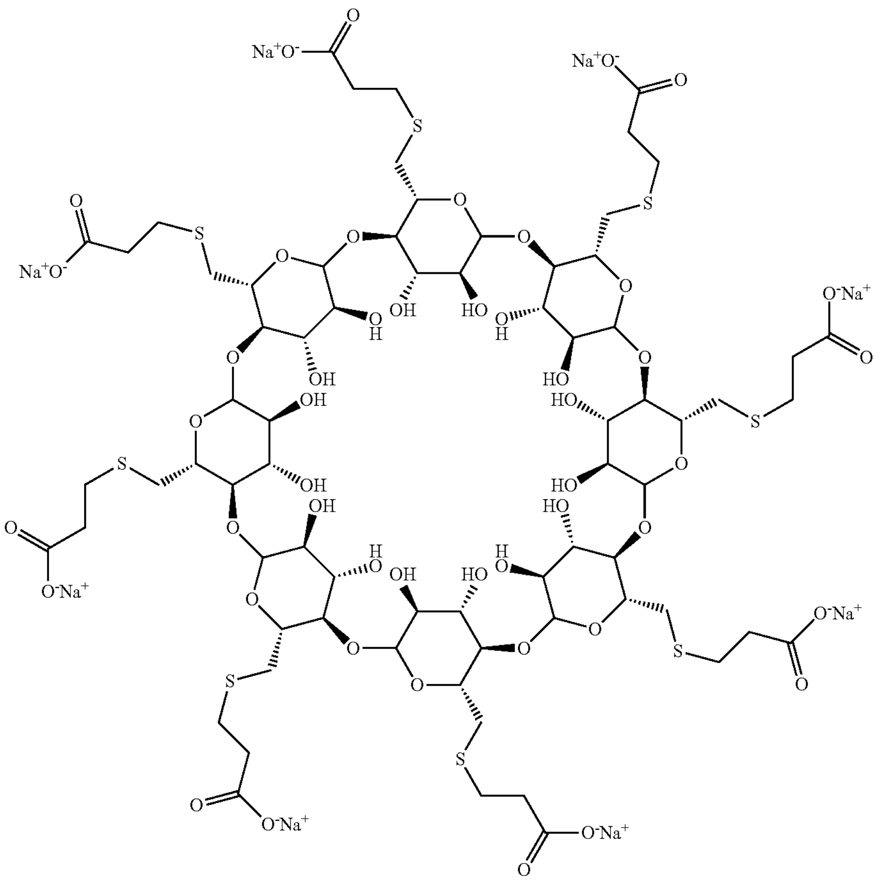
The present application provides an improved process for the preparation of sugammadex. More particularly, the application relates to a reaction of a salt of 3-mercapto propionic acid with 6-per-deoxy-6-per-chloro-γ-cyclodextrin to obtain sugammadex, wherein said salt is preferably selected from the group comprising of di sodium salt of 3-mercapto propionic acid, di potassium salt of 3-mercapto propionic acid and di lithium salt of 3-mercapto propionic acid. This application also relates to the use of isolated crystalline compound di alkali metal salt of 3-mercapto propionic acid for the preparation of sugammadex, which improves overall purity and/or reaction time of sugammadex.

The present application provides an improved industrially viable process, which is efficient, less time consuming, reproducible and involves less toxic reagents.

BACKGROUND OF THE INVENTION

Sugammadex is marketed as Bridion® and structurally known as compound of formula I, whereas formula I shows the sodium salt of sugammadex. It is an octa substituted γ-cyclodextrin derivative with a lipophilic core and a hydrophilic periphery.

Formula I



The chemical structure of cyclodextrins (CD) contains a cyclic oligosaccharides composed of α-(1,4) linked glucopyranose subunits. According to the general accepted nomenclature of cyclodextrins, an α (alpha)-cyclodextrin is a 6-membered ring molecule, a β (beta)-cyclodextrin is a 7-membered ring molecule and a γ (gamma)-cyclodextrin is a 8-membered ring molecule. Cyclodextrins are useful molecular chelating agents. They possess a cage-like supramolecular structure. As a result of molecular complexation phenomena CDs are widely used in many industrial products, technologies and analytical methods.

Sugammadex contains substituted γ-cyclodextrin with eight recurring amylose units each with five asymmetric carbon atoms, in total forty asymmetric carbon atoms for the whole molecule. The original configuration of all asymmetric carbon atoms is retained during the synthetic manufacturing process.

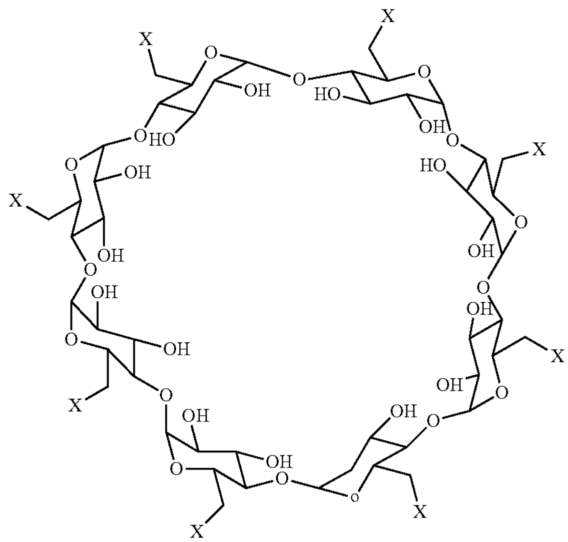
According to various disclosures, including disclosures associated with Bridion®, suganimadex is a potent and

SUMMARY OF THE INVENTION

In one aspect, the application provides an improved process for the preparation of sugammadex, said. process comprises the following steps;

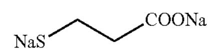
a) reacting 6-per-deoxy-6-per-halo-γ-cyclodextrin of a compound of formula II,

Formula II



wherein, X is chlorine, bromine or iodine with a salt of 3-mercapto propionic acid, selected from the group comprising of di sodium salt of 3-mercapto propionic acid, di potassium salt of 3-mercapto propionic acid and di lithium salt of 3-mercapto propionic acid, preferably with the di sodium salt of 3-mercapto propionic acid of formula III,

Formula III



in suitable organic solvent to obtain sugammadex,

b) optionally, purifying the sugammadex to obtain a sugammadex salt, preferably the sodium salt of formula I.

Formula I

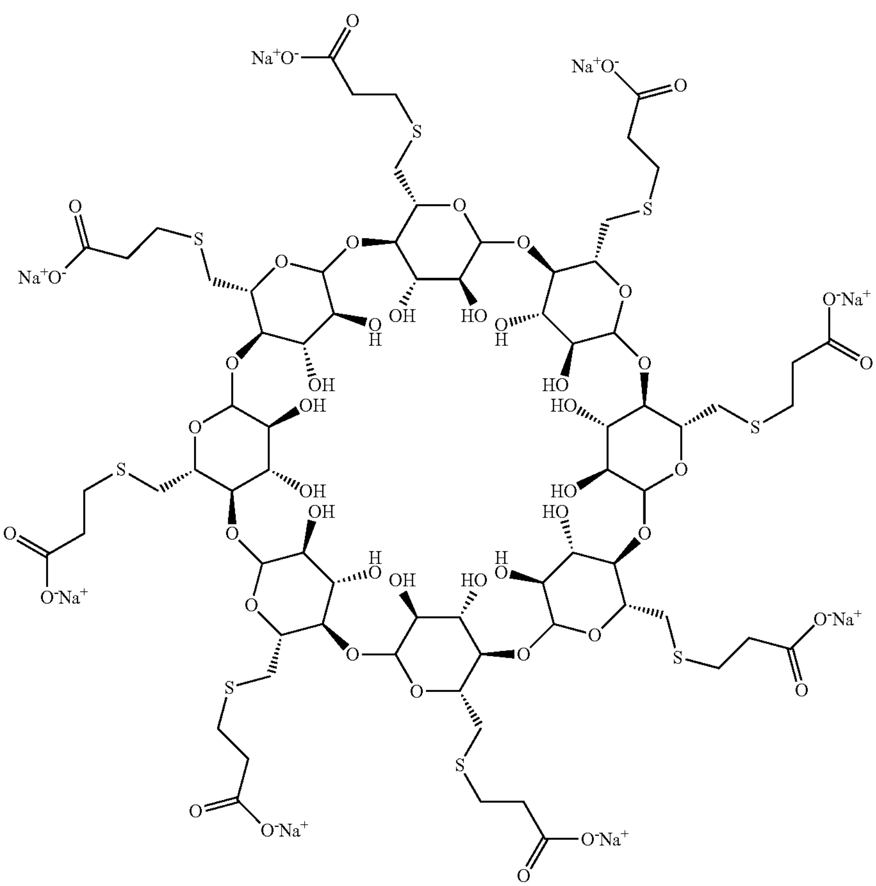


TABLE B-continued

Reference	Reaction time	Purity (area-% HPLC)
WO2012/025937A1 (Ref. Example 2)	12 Hours	94.35
WO2014/125501A1 (Ref. Example 3)	12-14 Hours	88.50
Example of the present application (Example 4C)	4-5 Hours	98.53
Example of the present application (Example 5)	4-5 Hours	98.28
Example of the present application (Example 6)	4-6 Hours	98.0

It is evident from the comparative data that process of present application requires lesser time for its completion and leads to higher purity of sugammadex.

According to the process of the present application, sugammadex is preferably prepared by reacting 6-per-deoxy-6-per-chloro-γ-cyclodextrin with isolated solid di sodium salt of 3-mercaptopropionic acid in suitable organic solvent. Said organic solvent can be selected form ethyl acetate, acetonitrile, propionitrile, tetrahydrofuran, dimethylformamide, dimethylacetamide, N-methyl-2-pyrrolidone, dimethyl sulfoxide or mixtures thereof. More preferably the organic solvent can be selected from dimethylformamide or dimethyl sulfoxide and most preferably the organic solvent is dimethyl sulfoxide.

The reaction can be carried out at any suitable temperature, but preferably from 40° C. to 90° C. or at the reflux temperature of the solvent. The reaction can be completed in about 2 to 6 Hours, more preferably in about 3 to 5 Hours.

Further, it is also advantageous to use dimethyl sulfoxide as reaction solvent in combination with the use of isolated solid di sodium salt of 3-mercaptopropionic acid, which accelerates the reaction leading to high substitutions in short reaction times and results in full substitutions of all eight halogenated positions of 6-per-deoxy-6-per-chloro-γ-cyclodextrin with 3-mercaptopropionic acid, more than 99% conversion is achieved by stirring the reaction mixture at 70° C. for about 2 to 6 Hours.

The completion of the reaction can be monitored by any suitable analytical technique. After completion of the reaction sugammadex may be isolated by any known methods which may include but not limited to cooling crystallization, anti-solvent addition, removal of solvent by evaporation, distillation, filtration of precipitated solid and the like; or any combinations of these methods.

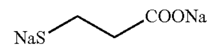
The isolated sugammadex may optionally be purified to achieve desired. purity and quality by techniques known in art like column chromatography, fractional distillation, acid base treatment, slurrying or recrystallization. Preferably, the sugammadex can be purified by dissolving it in a suitable solvent like water followed by mixing with an anti solvent to isolate sugammadex. It is also advantageous to treat the mixture with activated carbon prior to addition of an anti solvent. The anti solvent can be selected from alcoholic solvent such as methanol, ethanol, isopropyl alcohol, acetone, tetrahydrofuran, dimethylformamide and dimethyl sulfoxide or mixtures thereof.

The purified sugammadex may be optionally washed with suitable solvent and dried under suitable drying conditions. The drying may be suitably carried out using any of an air tray dryer, vacuum tray dryer, fluidized bed dryer, spin flash dryer, flash dryer, and the like. The drying may be carried out at any suitable temperatures and under atmospheric pressure or above, or under reduced pressures.

In another aspect, the present application provides the preparation and isolation of di alkali metal salt of 3-mercapto propionic acid. The di alkali metal salt of 3-mercapto propionic acid may be prepared by reacting 3-mercapto propionic acid with suitable base preferably an alkali metal hydroxide base in suitable organic solvent and isolating the di alkali metal salt of 3-mercapto propionic acid as solid. The alkali metal hydroxide base can be selected from lithium hydroxide, potassium hydroxide, sodium hydroxide or mixtures thereof The most preferred alkali metal hydroxide base is sodium hydroxide and di alkali metal salt of 3-mercapto propionic acid is di sodium salt of 3-mercapto propionic acid. The preferred organic solvent can be selected from the list of organic solvent as discussed above. The most preferred organic solvent is tetrahydrofuran.

In still yet another aspect, the application provides novel crystalline compound of di sodium salt of 3-mercapto propionic acid of formula III,

Formula III



and its use for the preparation of sugammadex.

The crystalline di sodium salt of 3-mercapto propionic acid of formula III shows on X-ray diffraction peak at an angle of refraction 2 theta (θ), of 9.08, 22.14, 23.76, 29.82 and 32.2 degrees; preferably it includes five or more peaks at angles of refraction 2 theta (θ) selected from the group consisting of 9.08, 11.50, 12.24, 14.47, 15.95, 16.33, 17.17, 18.18, 18.70, 20.09, 21.45, 21.96, 22.14, 23.07, 23.19, 23.76, 24.60, 25,48, 27.20, 27.42, 28.02, 28.31, 28.86, 28.98, 29.16, 29.82, 29.92, 30.44, 30.64, 31.14, 31.58, 31.78, 32.20, 32.46, 32.59, 32.98, 33.08, 33,97, 34.81, 35.96, 36.27, 36.98, 37.32, 37.43, 38.11, 38,55 and 38.66±0.2 degrees.

The 6-per-deoxy-6-per-halo-γ-cyclodextrin of compound, used as starting material for the preparation of sugammadex of the present application, can be prepared by any suitable methods known in the prior arts such as methods disclosed WO2001/040316A1, WO2012/025937A1 and WO2014/125501A1. Generally, it can be prepared by halogenation of γ-cyclodextrin with a suitable halogenating reagent in a suitable organic solvent. The suitable halogenating reagent can be selected from iodine, N-iodosuccinimide, oxalyl chloride, oxalyl bromide, thionyl chloride, thionyl bromide, phosphoryl chloride or phosphoryl bromide; more preferably the halogenating reagent can be selected from oxalyl chloride or thionyl chloride. The organic solvent for the halogenation reaction can be selected from the list of organic solvent as discussed above. The most preferred organic solvent is dimethylformamide.

The halogenation of γ-cyclodextrin produces a complex with the halogenating reagent and solvent, which is further hydrolysed with a suitable base in a suitable solvent/anti-solvent mixture to isolate 6-per-deoxy-6-per-halo-γ-cyclodextrin. Preferably the anti-solvent is a mixture of alcoholic solvent and water. The solvent of solvent:anti-solvent pair can be selected from the list of organic solvents as discussed above. The alcoholic component of the anti-solvent can be selected from C1-4 alcohol such as methanol, ethanol, propanol, isopropyl alcohol, n-butanol, iso-butanol, tert-butanol or mixtures thereof. The suitable base can be selected from alkali metal hydroxide such as sodium hydroxide, lithium, hydroxide, potassium hydroxide and the like; alkali metal carbonate and/or alkali metal bicarbonate such as potassium carbonate, lithium carbonate, sodium carbonate, potassium bicarbonate, lithium bicarbonate, sodium bicarbonate and the like.

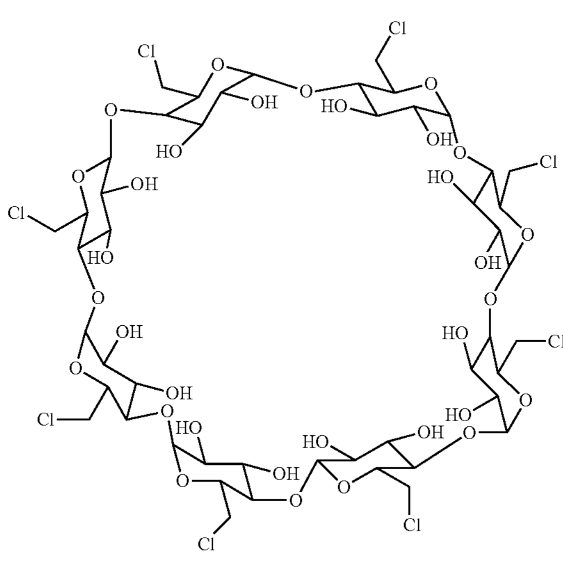
Preferably, the halogenation of γ-cyclodextrin is performed by reacting it with a suspension of oxalyl chloride or thionyl chloride in dimethylformamide under anhydrous condition at about −5° C. to 15° C., more preferably at about 0° C. to 10° C. The suspension of oxalyl chloride or thionyl chloride can be prepared by slow addition to dimethylformamide at about −5° C. to 15° C. The obtained suspension is further heated and mixed with γ-cyclodextrin. The mixing of γ-cyclodextrin with the suspension of oxalyl chloride or thionyl chloride can be achieved by addition of γ-cyclodextrin to the suspension or vice-versa, in both cases slow addition is preferred. After mixing, the obtained mixture may be stirred at a temperature of about 40° C. to 90° C., more preferably at about 60° C. to 80° C. for a period of about 5 to 25 Hours, preferably for about 10 to 20 Hours.

After completion of the reaction the solution is cooled and an alcoholic solvent, preferably methanol, is added then stirred for a period of time. The obtained solution is slowly added to an aqueous solution of base and methanol and then stirred for an appropriate time preferably for about 1 to 6 Hours, more preferably for about 1 to 4 Hours. The 6-per-deoxy-6-per-chloro-γ-cyclodextrin can be isolated from the reaction mixture by suitable techniques such as filtration, decantation or centrifugation and the like. The isolated 6-per-deoxy-6-per-chloro-γ-cyclodextrin may optionally be washed with aqueous methanol and if required, may optionally be further purified to obtain the desired purity of 6-per-deoxy-6-per-chloro-γ-cyclodextrin.

One embodiment of the process for preparing sugammadex comprises the following steps;

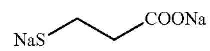
1) reacting 6-per-deoxy-6-per-chloro-γ-cyclodextrin of formula V,

Formula V



with isolated di sodium salt of 3-mercapto propionic acid of formula III,

Formula III



in dimethyl sulfoxide to obtain sugammadex,

2) dissolving sugammadex in water,
3) mixing with dimethyl sulfoxide,
4) stirring,
5) isolating sugammadex,
6) dissolving sugammadex of step 5) in water,
7) optionally, adding activated carbon,
8) mixing with methanol,
9) stirring,
10) isolating pure sugammadex.

Certain specific aspects of the present application will be explained in greater detail with reference to the following examples, which are provided only for purposes of illustration and should not be construed as limiting the scope of the disclosure in any manner.

EXAMPLES

To demonstrate the benefits of the present invention, example of the prior art were worked and indicated as reference example.

Reference Example 1 (Example 4 of WO2001/040316A1)

Preparation of Sugammadex

3-Mercapto propionic acid (1.95 mL, 10 eq,) was dissolved in dry dimethylformamide (72 mL) under nitrogen at room temperature. To this solution was added in three portions sodium hydride (1.97 g, 22 eq., 60% dispersion in mineral oil) and the mixture was stirred for another 30 minutes. To this mixture was then added drop wise a solution of 6-per-deoxy-6-per-iodo-γ-cyclodextrin (5 g, 2.24 mmol) in dry dimethylformamide (72 mL). After addition, the reaction mixture was heated at 70° C. for 12 Hours. After cooling, water (16 mL) was added to the mixture and the volume was reduced to 64 mL in vacuo. Ethanol (400 mL) was added resulting in precipitation. The solid precipitate was collected by filtration and dialyzed for 36 Hours. The volume was then reduced to 32 mL in vacuo. To this ethanol was added, and the precipitate was collected by filtration and dried to give the sugammadex as a white solid (2.1 g, 42% by wt) with a purity of 88.75 area-% HPLC.

Reference Example 2 (Example 2 of WO2012/025937A1)

Preparation of Sugammadex

To a mixture of sodium hydride (60% dispersion in mineral oil, 3.13 g) in dimethylformamide (19 mL, 3.8 vol.), a solution of 3-mercapto propionic acid (3 mL, 10 eq.) in dimethylformamide (6 mL, 1.2 vol.) was added slowly under nitrogen maintaining the temperature below 10° C. The resulting mixture was stirred at 20-25° C. for 30 minutes. Then 6-per-deoxy-6-per-chloro-γ-cyclodextrin (5 g) in dimethylformamide (50 mL, 10 vol.) was added slowly at 5-10° C. under nitrogen and the resulting mixture was heated to 70-75° C. for 12 Hours. Reaction mixture was cooled to 20-25° C. and dimethylformamide removed partially under vacuum and the reaction mixture is diluted with reaction, the reaction mass was slowly cooled to 20-30° C., then filtered under nitrogen, washed with dimethyl sulfoxide (100 ml), ethanol (750 ml) and dried under vacuum at 40-50° C. for 15 Hours to give 43.7 g crude sugammadex. ^1H NMR (400 MHz, D_2O): δ 2.53-2.60 (m, 16H); δ 2.89-2.93 (t, 16H); δ 3.02-3.07 (m, 8H); δ 3.16-3.19 (d, 8H); δ 3.67-3.73 (m, 16H); δ 3.98-4.03 (t, 8H); δ 4.10-4.13 (m, 8H); δ 5.24-5.25 (d, 8H).

Example 4B: The crude sugammadex (43 g) was added to 180 ml of water at 20-30° C. and the mixture was stirred for 10-20 minutes at the same temperature to get a clear solution. Dimethyl sulfoxide (205 ml) was slowly added to the reaction mixture at 25-35° C. and the reaction mixture was then stirred at 20-30° C. for 1 hour. Further Dimethyl sulfoxide (65 mL) was added and the reaction mixture stirred for 1 hour at same temperature. The resultant solid was filtered, washed with 10% aqueous dimethyl sulfoxide (100 ml), ethanol (300 ml) and dried at 50-55° C. for 12 hour to give 28.10 g sugammadex.

Example 4C: The sugammadex (28 g) obtained from example 4B was dissolved in water (380 ml) and methanol (380 ml) at 20-30° C. To this solution activated carbon (2.8 g, 10% w/w) was added and stirred for 2 Hours at 20-30° C. The reaction mixture was filtered through celite and washed with methanol:water (1:1) (38 ml). The filtered solution was heated to 45-50° C. and methanol (2400 ml) was slowly added at the same temperature. The reaction mixture was then slowly cooled to 20-30° C. and stirred at the same temperature for 1 Hour. The reaction mass was filtered, washed with methanol (100 ml) and dried under vacuum at 60-65° C. for 15 Hours to give 24 g of sugammadex. The product had purity of 98.53 area-% HPLC, Impurity I=Not Detected, Impurity II=Not Detected, Impurity III=0.45.

Example 5

Preparation of Sugammadex

Dimethyl sulfoxide (200 ml) was de-oxygenated with nitrogen, vacuum and nitrogen at 20-30° C. The di-sodium salt of 3-mercaptopropionic acid (16.6 g, 0.1107 moles) was added and the reaction mass was again de-oxygenated with nitrogen and vacuum at 20-25° C. The 6-per-deoxy-6-per-chloro-γ-cyclodextrin (10 g, 0.0069 moles) was dissolved in dimethyl sulfoxide (50 ml) under nitrogen at 20-30° C. and added to the reaction mass at 20-25° C. Then the reaction mixture was de-oxygenated with vacuum and nitrogen at 20-30° C. The reaction mixture was heated to 70-75° C. and stirred for 4-5 hour at 70-75° C. The reaction mass was cooled to 20-30° C. and diluted with water (250 ml) at 20-35° C. Slowly added dimethyl sulfoxide (180 ml) to the reaction mass in two lots at 20-35° C. and stirred for 1 hour at 20-30° C., then filtered the mass under nitrogen, washed with 20% water in dimethyl sulfoxide (40 ml), suck dried under vacuum 1-2 hours to give Sugammadex with purity 98.28%, Impurity I=Not Detected, Impurity II=Not Detected, Impurity III=0.24.

^1H NMR (400 MHz, D_2O): δ 2.53-2.60 (m, 16H); δ 2.89-2.93 (t, 16H); δ 3.02-3.07 (m, 8H); δ 3.16-3.19 (d, 8H); δ 3.67-3.73 (m, 16H); δ 3.98-4.03 (t, 8H); δ 4.10-4.13 (m, 8H); δ 5.24-5.25 (d, 8H).

Example 6

Preparation of Sugammadex

Dimethyl sulfoxide (2000 ml) was de-oxygenated with nitrogen, vacuum and nitrogen at 20-30° C., The di-sodium salt of 3-mercaptopropionic acid (166.2 g, 1.107 moles) was added and the reaction mass was again de-oxygenated with nitrogen and vacuum at 20-25° C. The 6-per-deoxy-6-per-chloro-γ-cyclodextrin (100 g, 0.069 moles) was dissolved, in dimethyl sulfoxide (500 ml) under nitrogen at 20-30° C. and added to the reaction mass at 20-25° C. Then the reaction mixture was de-oxygenated with vacuum and nitrogen at 20-30° C. The reaction mixture was heated to 70-75° C. and stirred for 4-6 hour at 70-75° C. The reaction mass was cooled to 20-30° C. and diluted with water (2500 ml) at 20-35° C. Slowly added dimethyl sulfoxide (1800 ml) to the reaction mass in two lots at 20-35° C. and stirred for overnight at 20-30° C., then filtered the mass under nitrogen, washed with 20% aqueous dimethyl sulfoxide (400 ml), suck dried under vacuum 1-2 hours to give 220 gm Sugammadex.

The obtained Sugammadex was dissolved in 800 ml of water. Dimethyl sulfoxide (1600 ml) was slowly added to the reaction mixture at 25-35° C. in two lots and the mass was stirred for 3 hour at 20-30° C. The mass was filtered and washed with 20% aqueous dimethyl sulfoxide (400 ml), methanol (500 ml) and suck dried under nitrogen for 1 hour to give 100 g wet Sugammadex.

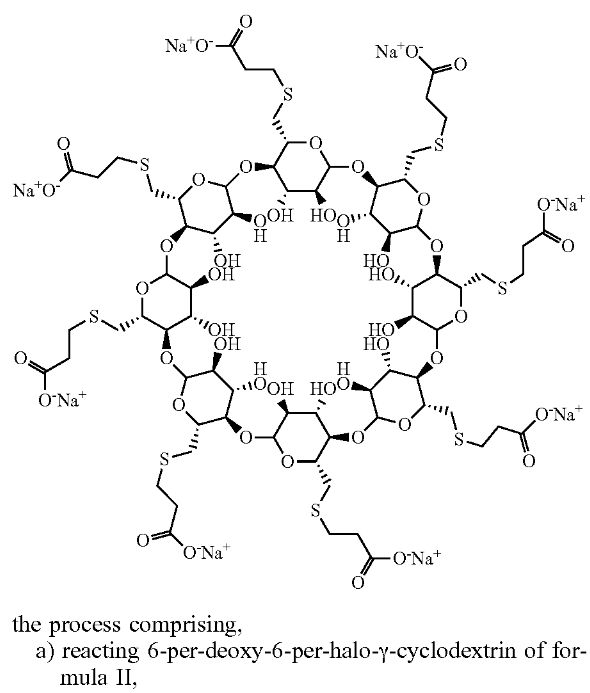
The obtained solid was dissolved in water (500 ml). To this solution activated carbon (10 g) was added and stirred for 1-2 Hours at 20-30° C. The mass was filtered through 5 micron and then washed with water (300 ml). The filtered mass was diluted slowly by adding methanol (4800 ml) at 20-35° C. The mass was stirred at 20-30° C. for 5-6 Hours. The mass was filtered, washed with methanol (400 ml) and dried under vacuum at 55-65° C. for 12-15 hours to give 70 g of Sugammadex. The obtained Sugammadex (70 g) was dissolved in water (500 ml), treated with activated carbon and filtered through 0.45 micron, The filtrate was lyophilized to give 60 g of Sugammadex. The product had purity of 98.0 area-% HPLC, Impurity I=Not Detected, Impurity II=Not Detected, Impurity III=0.37.

^1H NMR (400 MHz, D_2O): δ 2.53-2.60 (m, 16H); δ 2.89-2.93 (t, 16H); δ 3.02-3.07 (m, 8H); δ 3.16-3.19 (d, 8H); δ 3.67-3.73 (m, 16H); δ 3.98-4.03 (t, 8H); δ 4.10-4.13 (m, 8H); δ 5.24-5.25 (d, 8H).

The invention claimed is:

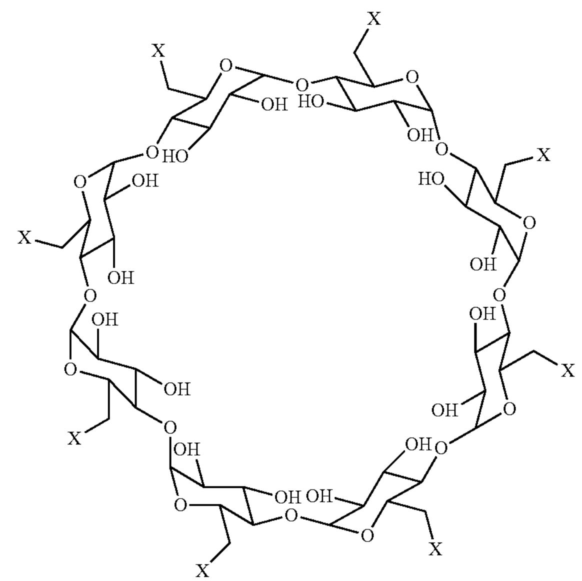
1. A process for preparing sugammadex of formula I,

Formula I



the process comprising,

a) reacting 6-per-deoxy-6-per-halo-γ-cyclodextrin of formula II, Formula II



wherein, X is chlorine, bromine or iodine,

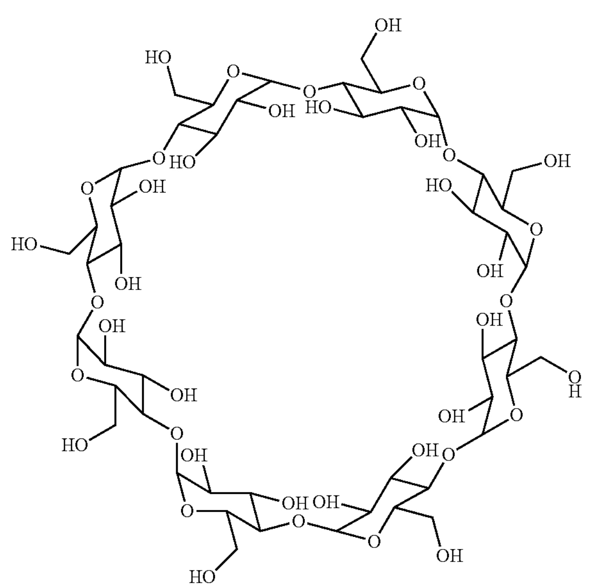
with a salt of 3-mercapto propionic acid, selected from di sodium salt of 3-mercapto propionic acid, di lithium salt of 3-mercapto propionic acid, and di potassium salt of 3-mercapto propionic acid, in an organic solvent to obtain sugammadex, wherein the salt is isolated, and

b) optionally, purifying the sugammadex.

2. The process according to claim **1** wherein the 6-per-deoxy-6-per-halo-γ-cyclodextrin is prepared by a process that comprises,

i) reacting γ-cyclodextrin of formula IV,

Formula IV



with a halogenating agent, in an organic solvent,

ii) adding to the reaction mixture of step i) an aqueous solution of base and alcoholic solvent,

iii) isolating the 6-per-deoxy-6-per-halo-γ-cyclodextrin of formula II,

iv) optionally, drying the 6-per-deoxy-6-per-halo-γ-cyclodextrin, and

v) optionally, purifying the compound of step iii) or iv).

3. The process according to claim **1**, wherein the salt of 3-mercapto propionic acid is the di sodium salt of 3-mercapto propionic acid.

4. The process according to claim **3**, wherein the di sodium salt of 3-mercapto propionic acid is prepared by a process that comprises,

i) reacting 3-mercaptopropionic acid with an alkali metal hydroxide base in an organic solvent, to produce the di sodium salt of 3-mercapto propionic acid,

ii) isolating the di sodium salt of 3-mercapto propionic acid, and

iii) optionally, drying the di sodium salt of 3-mercapto propionic acid.

5. The process according to claim **1**, wherein the organic solvent is selected from ethyl acetate, acetonitrile, propionitrile, tetrahydrofuran, dimethylformamide, dimethylacetamide, N-methyl-2-pyrrolidone, dimethyl sulfoxide and mixtures thereof.

6. The process according to claim **2**, wherein the halogenating agent is iodine, N-iodosuccinimide, oxalyl chloride, oxalyl bromide, thionyl chloride, thionyl bromide, phosphoryl chloride or phosphoryl bromide.

7. The process according to claim **2**, wherein the base used in step ii) is selected from alkali metal hydroxides, alkali metal carbonates, alkali metal bicarbonates, and mixtures thereof.

8. The process according to claim **2**, wherein the alcoholic solvent of step ii) is selected from methanol, ethanol, propanol, isopropyl alcohol, n-butanol, isobutanol, tert-butanol and mixtures thereof.

9. The process according to claim **4**, wherein the alkali metal hydroxide base of step i) is selected from lithium hydroxide, potassium hydroxide, sodium hydroxide and mixtures thereof.

10. The process according to claim **1**, wherein the di sodium salt of 3-mercapto propionic acid is isolated as a solid before its reaction with the 6-per-deoxy-6-per-halo-γ-cyclodextrin of formula II.

11. The process according to claim **1**, comprising purifying sugammadex according to step b) by a process that comprises,

b1) dissolving sugammadex in solvent,

b2) optionally, adding activated carbon,

b3) adding an anti solvent,

b4) stirring, and

b5) isolating purified sugammadex.

12. The process according to claim **11**, wherein solvent of step b1) is water.

13. The process according to claim **11**, wherein anti solvent of step b3) is selected from methanol, ethanol, isopropyl alcohol, acetone, tetrahydrofuran, dimethylformamide and dimethyl sulfoxide.

14. The process according to claim **1**, wherein the organic solvent of step a) is dimethyl sulfoxide.

15. The process according to claim **1**, comprising,

1) reacting 6-per-deoxy-6-per-chloro-γ-cyclodextrin of formula V